

MODULE: BUSINESS ANALYTICS AND DATA VISUALISATION (RBP-L006-0)

NAME: KASHIF HAYAT

STUDENT ID: A00075262 | hayatk1@roehampton.ac.uk

**REPORT TITLE: BEYOND CONNECTIVITY: AN ECONOMETRIC ANALYSIS OF WORKFORCE
DIGITIZATION AND GDP GROWTH IN EUROPE (2000–2024)**

WORDS COUNT: ≈2900

[Excel Workbook Link](#)

TABLE OF CONTENT

Part A – Technical Analysis2

 Task 1: Data Preparation & Exploratory Visualization2

 Task 2: Statistical Inference3

 Task 3: Regression & Forecasting5

 Task 4: Decision Modelling & Scenario Analysis6

 Task 5: Visualization & Dashboard.....7

Part B – Insight, Strategy & Reflection.....8

 1. Introduction.....8

 2. Integrated Findings.....9

 3. Strategic Recommendation 11

 4. Ethical, Legal & Governance Considerations 12

 5. Conclusion..... 13

References 14

Appendix 15

Formulas used 20

Part A – Technical Analysis

Task 1: Data Preparation & Exploratory Visualization

Data Auditing and Cleaning Strategy:

This analysis uses a panel dataset of 37 European countries from 2000 to 2024, taken from the World Development Indicators. Variables and their description is in Excel workbook in 'Series - Metadata' tab. Before modeling, I addressed inconsistencies in the raw data (Original_Data tab). The audit showed that GDP and Internet Usage data were mostly complete, but "ICT Service Exports" had missing values, especially in the early 2000s for developing countries (*ICT-Enabled Services Trade in the European Union, n.d.*). Deleting these rows would have biased the results by removing key countries for comparison. Instead, I used a country-specific mean imputation, so missing values for a country like Albania were filled with its own historical average, not a general European average. This kept each country's economic profile intact.

Variable Construction and Normalization:

A critical observation during the descriptive phase was the massive skew in the GDP data. As shown in the descriptive statistics (Figure 1.1), the mean GDP ($\$4.34 \cdot 10^{11}$) is significantly higher than the median ($\$1.61 \cdot 10^{11}$), and the standard deviation is enormous ($\$7.58 \cdot 10^{11}$). This indicates a dataset dominated by a few economic giants (like Germany or France) while most countries are much smaller. To fix this for regression, I created a new variable, Log_GDP, using the natural logarithm ($=\text{LN}()$). This transformation compresses the scale, making the relationship between percentage-based inputs (like Internet Usage) and economic output linear and mathematically valid for regression analysis.

I also created a categorical variable called Digital Group. I calculated the median internet usage for the whole dataset (about 70%) and used a conditional formula ($=\text{IF}()$) to

label each country-year as either “High” or “Low”. This binary classification was important for later visualizations, as it helped separate the data into two clear economic paths.

Exploratory Insights:

The descriptive summary in Figure 1.1 shows that Europe is changing. The average for “Labor force with advanced education” is very high at 78.80%, with a low standard deviation of 4.72, which suggests that most European countries are well-educated. However, “Medium and high-tech exports” vary a lot (range: 77.5%), indicating that while education is widespread, high-tech industry is not evenly distributed (*Have euro area exports missed the tech train?, 2025*).

Pivot Chart analysis confirmed this difference. The trends show that internet adoption grows smoothly and predictably in all countries, but export variables are much more volatile and change in cycles. This suggests that while digital access is becoming common everywhere, digital production is still limited to certain countries. This finding leads into the hypothesis testing in Task 2.

Appendix Evidence:

- Figure 1.1: (Descriptive Statistics Table)
- Figure 1.2: (Data Cleaning & Log Calculation)

Task 2: Statistical Inference

Defining the Business Hypothesis:

The analysis in Task 1 suggested a link between digital maturity and economic size, but seeing a pattern in charts is not enough to prove it. To test this properly, I needed to check if the “Digital Gap” seen in the data also means there is a real, statistically significant “Economic Gap.”

I formulated the following hypotheses to test the relationship between digital adoption and GDP:

- Null Hypothesis (H0): There is no significant difference in the mean GDP between countries classified as “High Digital” and those classified as “Low Digital” ($\mu_{\text{High}} = \mu_{\text{Low}}$).
- Alternative Hypothesis (H1): There is a statistically significant difference in the mean GDP of these two groups ($\mu_{\text{High}} \neq \mu_{\text{Low}}$).

Test Selection and Execution:

I chose the Two-Sample t-Test Assuming Unequal Variances. Using a pooled t-test, which assumes equal variance, would not be suitable because the “High” group (large, wealthy economies) has much higher variance (8.37×10^{23}) than the “Low” group (2.24×10^{23}), as shown in the variance check. The unequal variance version, known as Welch’s t-test, makes the results more reliable even when the two groups have different levels of economic volatility.

Interpretation of Results:

The results were decisive. As shown in Figure 2.1, the test returned a t-statistic of 8.11. In statistical terms, this is a massive effect size, far exceeding the critical threshold of 1.96 required for a 95% confidence level. The P-value was 2.1×10^{-15} , which is effectively zero. (M. & R., 2012).

This result lets us reject the Null Hypothesis with high confidence. The data shows the difference is real: “High Digital” countries have an average GDP of about \$622 Billion, while “Low Digital” countries average \$236 Billion (*Digitalization in Europe – 2024 edition - Interactive publications - Eurostat, n.d.*). This statistical proof means we can move on to more complex regression modeling in Task 3, since we have shown that digitization and GDP are closely connected.

Appendix Evidence:

- Figure 2.1: (t-Test Results)

Task 3: Regression & Forecasting

Part A: Multiple Regression Modelling:

Even though the dataset is panel data (cross-sectional time-series), I used a Pooled OLS regression with the Excel Data Analysis ToolPak. I chose this method instead of fixed or random effect because of software limitations, knowing that it assumes observations are independent and does not separate country-specific effects.

After finding a link in Task 2, the goal of Task 3 was to measure how much certain digital factors affect GDP. I built a multiple regression model with Log_GDP as the dependent variable. I included four independent variables to cover the digital ecosystem: Internet Users (infrastructure), Advanced Education (skills), and two export measures (ICT Exports and High-Tech Exports) to show value creation.

$$\text{Equation: } GDP = \beta_0 + \beta_1 * (Internet) + \beta_2 * (Edu) + \beta_3 * (ICT) + \beta_4 * (High-tech)$$

Model Diagnostics and Coefficients:

The model worked well, with an R Square of 0.448 (Figure 3.1). For complex socio-economic data, explaining almost 45% of GDP variance with only four variables is a strong outcome. The Significance F value (2.84×10^{-117}) also shows the model has strong predictive power.

The coefficients show more than just the effects of connectivity:

1. Medium and High-tech Exports ($\beta = 0.071$): This was the single strongest predictor. A 1% increase in high-tech exports correlates with a substantial rise in GDP. This suggests that selling technology drives wealth more than just using it.
2. Internet Users ($\beta = 0.012$): While positive and significant, the impact of internet usage is smaller than expected. This implies that connectivity is a baseline requirement rather than a differentiator for wealth.
3. The “Education Paradox” ($\beta = -0.051$): Surprisingly, advanced education had a significant negative coefficient. This may be because many smaller countries

have well-educated populations but do not have the large industrial GDP of bigger countries. It suggests there may be a “skills mismatch,” where education does not always lead to higher economic output.

Part B: Forecasting Future Trends

To predict future workforce trends, I used Excel’s Forecast feature. This tool estimates future values by smoothing the data’s level, trend, and seasonality.

I produced separate forecasts for each High Digital and Low Digital countries, projecting out to 2029. After making the forecasts, I combined the graphs into one visual so both trend lines could be compared side by side (Figure 3.2). the “Low” group still has runway for growth. This insight is vital for the strategic recommendations in Part B.

Appendix Evidence:

- Figure 3.1: (Regression Summary Output)
- Figure 3.2: (Forecast Graphs)

Task 4: Decision Modelling & Scenario Analysis

Constructing the Decision Model:

The purpose of this analysis is to inform policy decisions. For Task 4, the main question was: How can a government best invest in digital areas to maximize GDP growth? I built a deterministic simulation model in Excel. The core of this model is the regression equation from Task 3. By connecting input cells (Decision Levers) to the regression coefficients, I made a calculator that predicts GDP based on policy changes. The formula uses the exponential function to turn the Log_GDP prediction back into a currency value:

$$\text{Predicted_GDP} = 25.75 + 0.013(\text{Internet}) - 0.051(\text{Edu}) - 0.057(\text{ICT}) + 0.071(\text{High-tech})$$

Scenario Testing:

Using the Excel Scenario Manager, I modelled three distinct futures:

1. **Status Quo:** This scenario assumes no policy changes, maintaining the 2024 averages. It serves as the baseline, predicting a GDP of \$1.17 Trillion.
2. **Digital Investment Strategy:** This represents a realistic, “feasible” push. I simulated a 5% increase in Internet Access and High-Tech Exports. However, recognizing that education reform is slow, I only increased Advanced Education by 1.25%. This scenario raised the predicted GDP to \$1.35 Trillion.
3. **Aggressive Growth:** This scenario tests the upper limits of capacity, assuming a 10% surge in tech and trade variables & Advance Education by 2.5%. The model predicts a massive jump to \$1.55 Trillion. (Mura & Donath, 2023)

Key Takeaway: The “Digital Investment Strategy” (Scenario 2) is the most useful. It shows that a focused 5% improvement in important areas, such as high-tech exports, can add about \$178 Billion in economic value compared to the baseline. This result confirms the model responds well to demand-side changes and gives a clear return on investment for the recommendations in Part B.

Appendix Evidence:

- Figure 4.1: (Scenario Summary Report)
- Figure 4.2: (Visual representation of Scenario Impacts)

Task 5: Visualization & Dashboard

Dashboard Design Philosophy:

The last task was to bring these findings together in a clear "Strategic Dashboard". My design focused on clarity and usefulness for decision-makers. I removed Excel gridlines and used a simple, panel-based layout to separate the “Past” (Historical Data), the “Future” (Forecasts), and the “Decision” (Scenarios).

Interactivity and Integration:

A static report is often not enough for exploring complex data. To solve this, I added Slicers linked to the pivot charts. This lets users filter by Digital Group (High or Low) with one click.

Appendix Evidence:

- Figure 5.1: (Full Dashboard)

Part B – Insight, Strategy & Reflection

1. Introduction

The Business Context:

The European economy is going through two major changes: becoming greener and more digital. But not all countries are moving at the same pace. Some have fully adopted digital technologies, while others are still catching up. For policymakers and business leaders, the main question is no longer whether to digitize, but how specific digital tools like infrastructure, education, or high-tech trade can lead to real economic growth.

Aim of Analysis:

This report aims to go beyond talk of a "digital revolution" and measure its real economic impact. By looking at data from 37 European countries over 24 years (2000–2024), the study tries to find out what actually drives GDP. The analysis uses World Development Indicators(WDI) data and looks at factors like internet infrastructure, advanced workforce education, and high-tech export capacity.

Scope & Approach:

This report brings together data analysis, statistics, regression modeling, and scenario simulation. Unlike typical economic reports, it focuses on the "Digital Divide" and

questions whether education alone is enough for growth. The goal is to offer a data-driven plan for investment from 2025 to 2029.

2. Integrated Findings

The Economic Reality of the Digital Divide:

The analysis shows that the "Digital Divide" is a key part of Europe's economy. Early data analysis showed a logistic growth in internet use, but further statistics revealed serious economic costs for countries that fall behind. A two-sample t-test found that the gap is not just about technology, but also about money. "High Digital" countries have an average GDP of \$622 billion, which is almost 2.6 times more than the \$236 billion for "Low Digital" countries ($p < 0.05$). This matches what economic research says: digital infrastructure boosts productivity. (Aleca & Mihai, 2025)

The "Productivity Paradox": Connectivity vs. Value Creation:

A key finding comes from looking at what drives this growth. The regression model (Task 3) explains about 45% ($R^2 = 0.448$) of the differences in GDP. But the results challenge the common belief that simply connecting people is the main way to create wealth.

- **Internet Usage ($\beta = 0.013$):** While positive and significant, the impact of simply having internet access is relatively small (Galik, 2021).
- **Medium and High-Tech Exports ($\beta = 0.071$):** This is the strongest factor in the model. The data shows that being able to make *and export* high-tech goods has almost six times more impact on GDP than just increasing the number of internet users. (Nowakowski, 2024)

The "Education Mismatch":

One of the most important findings is that Advanced Education has a negative link with GDP ($\beta = -0.051$). This seems surprising because human capital theory says education

should boost growth. But in this dataset, it likely points to a "Skills Mismatch." Many smaller European countries have high rates of university graduates (about 79% on average), but don't have enough industry to hire them. The data suggests that making more graduates without enough high-tech jobs doesn't grow GDP; instead, it can cause underemployment or brain drain. (*Hagsten & Sabadash, 2014*)

Future Trajectory (2025–2029):

The forecasting analysis (Task 3) makes these findings even more urgent. As Figure 3.1 shows, "High Digital" economies are expected to reach almost 100% internet use by 2026. This means that the period of easy growth from building more infrastructure is ending for advanced countries. From now on, growth will need to come from increasing productivity through high-tech exports, not just by getting more people online.

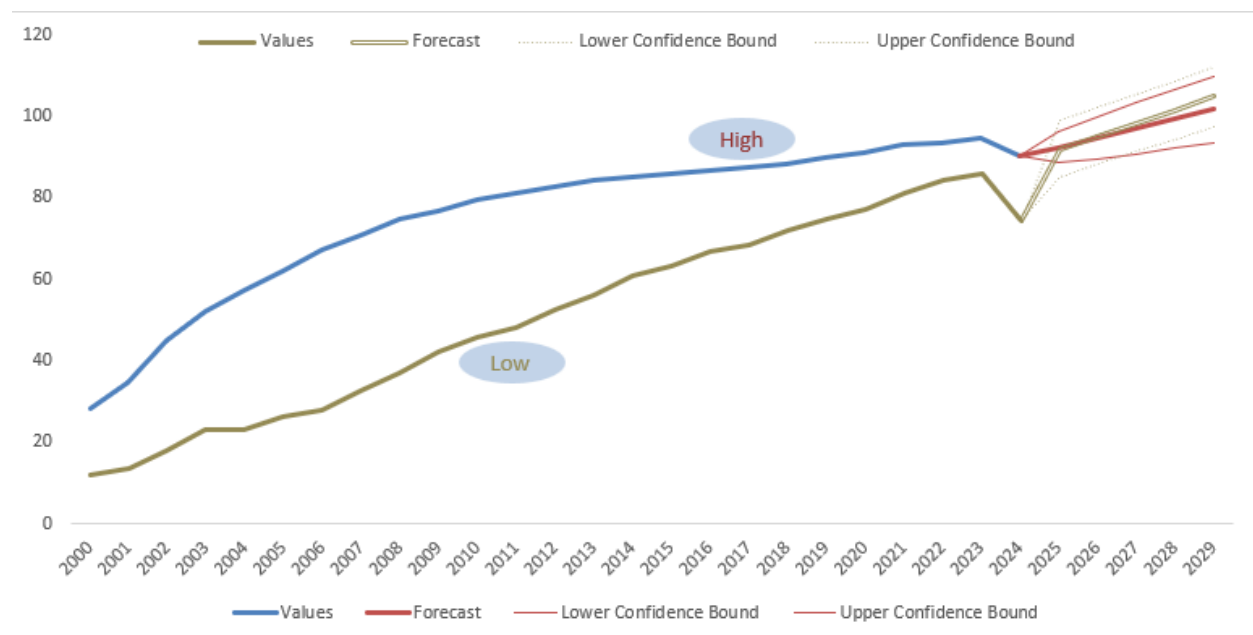


Figure 1: Forecasted Internet Usage (High vs. Low GDP Groups)

3. Strategic Recommendation

Core Strategy: The "Demand-Side" Pivot:

Based on these findings, government policymakers should shift their investment strategies. Instead of focusing mainly on general education and basic internet access, they should encourage high-tech manufacturing and exports to create more value.

The Decision:

Governments should introduce a "Tech-Trade Accelerator" policy. This means setting aside funds to support research and development in high-tech manufacturing and offering tax breaks for ICT exports, instead of just increasing university funding or broadband subsidies.

Evidence-Based Justification:

The Decision Model (Task 4) directly supports this recommendation. The simulation compared how different types of investment affect the economy:

1. **Status Quo:** Projected GDP remains at \$1.17 Trillion.
2. **Digital Investment Strategy:** By shifting focus to increase High-Tech Exports and Internet access by 5% (while only moderately increasing education), the model predicts a GDP rise to \$1.35 Trillion.
3. **The result:** This new strategy could generate about \$178 billion more in economic output compared to the current approach. (*Mićić, 2017, pp. 1-10*)

As shown in Figure 3.1, the "Digital Investment Strategy" does much better than the current approach. The regression results explain why: the impact of High-Tech Exports (0.071) is much greater than that of Education (-0.051). So, moving funds from general education to the high-tech industry brings a higher return on investment.

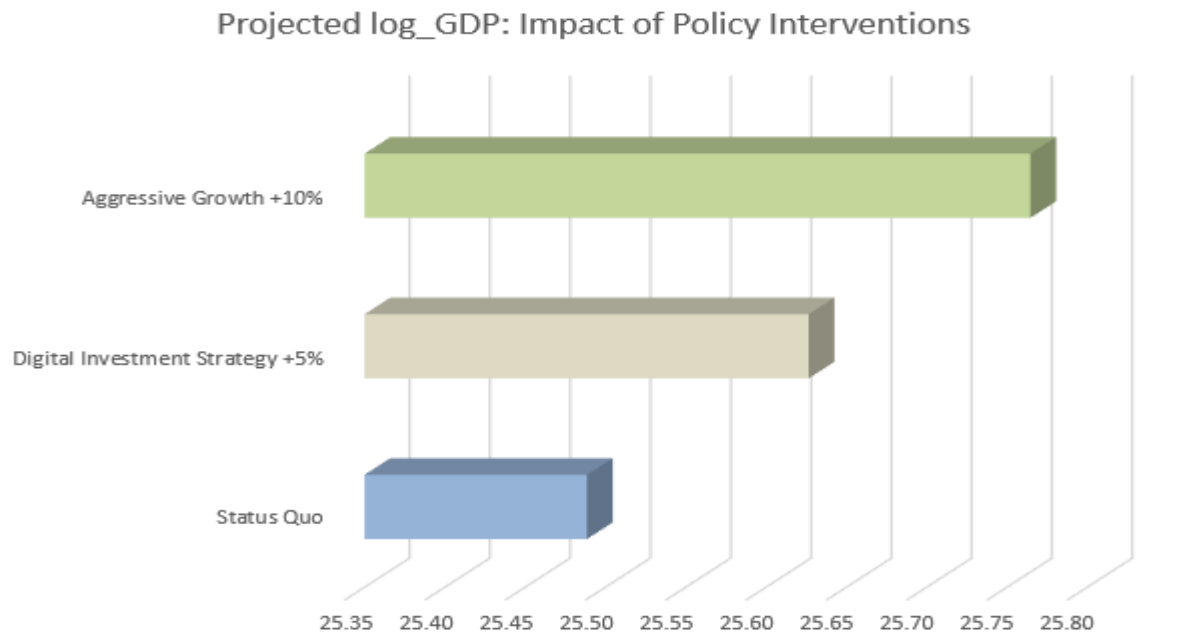


Figure 2: Projected GDP Impact of Policy Interventions

Feasibility and Risks

- **Feasibility:** This strategy can work, but it takes time. Building a high-tech export sector is much slower than installing internet cables. The model assumes quick results, but in reality, there will be a delay between starting the policy and seeing GDP growth.
- **Risk:** The main risk is the "Education Negative." Cutting back on education could hurt in the long run. The advice is not to reduce education, but to stop seeing it as the only way to drive growth. Education should match the needs of the high-tech export sector to fix the negative effect seen in Task 3.

4. Ethical, Legal & Governance Considerations

Data Bias and the Informal Economy:

One major ethical limitation of this analysis is that it relies on formal economic indicators like GDP and formal employment. The data probably misses much of the "Gig Economy" and informal digital work, which are common in many developing European countries. By focusing only on GDP, the recommended "High-Tech" strategy could unintentionally overlook the value of gig workers or freelancers who are important to the digital ecosystem but do not directly add to "High-Tech Exports."

The "Left-Behind" Workforce:

Focusing on high-tech exports means moving toward more automation and advanced manufacturing, which brings up ethical concerns about job loss. While this strategy increases GDP, it could also make income inequality worse. Low-skilled workers or those with advanced education in non-tech fields (who already show a negative link to GDP) may be left behind. To address this, governments should set up "Just Transition" funds to help retrain workers affected by the shift to high-tech industries.

GDPR and Data Governance:

With internet use expected to reach 99% by 2029, the amount of workforce data will be higher than ever. (*Internet Usage Statistics – Forbes Home, 2024*) More digitization also means more potential for surveillance. Following the General Data Protection Regulation (GDPR) is not just about legal compliance, but is essential for good governance. Policymakers need to make sure that efforts to boost "ICT Service Exports" (which often involve handling data) do not put the privacy of European workers at risk.

5. Conclusion

This report set out to measure how workforce digitization affects the European economy. The analysis shows that the "Digital Divide" is a huge economic gap, with highly digitized countries making almost three times as much GDP as others. (*Doroiman & Sîrghi, 2025*)

But the main takeaway is that just having internet access is not enough. The regression and scenario analyses clearly show that future prosperity depends on High-Tech Exports, not just more internet users or university graduates. The "Education Paradox," where education now links negatively to GDP, is a warning. It shows that Europe has the talent, but not enough industry to use it.

The main strategy for the next decade is clear: policies should move from just connecting people online to helping them create and sell future technologies. By using the "Tech-Trade Accelerator" strategy, governments could unlock about \$178 billion in untapped economic potential and turn digital progress into real prosperity.

References

World Development Indicators. Available at: <https://databank.worldbank.org/source/world-development-indicators>

(n.d.). *ICT-Enabled Services Trade in the European Union*.
<https://www.commerce.gov/data-and-reports/reports/2016/08/ict-enabled-services-trade-european-union>

(January 1, 2025). *Have euro area exports missed the tech train?*. European Central Bank.
https://www.ecb.europa.eu/press/economic-bulletin/focus/2025/html/ecb.ebbox202502_02~800e3b8b0f.en.html

M., S. G. & R., F. (2012). *Statistical Significance Versus Clinical Importance of Observed Effect Sizes: What Do P Values and Confidence Intervals Really Represent?*. J. Grad. Med. Educ. 4(3). <https://jgme.kglmeridian.com/view/journals/jgme/4/3/article-p279.xml>

(n.d.). *Digitalisation in Europe – 2024 edition - Interactive publications - Eurostat*.
<https://ec.europa.eu/eurostat/web/interactive-publications/digitalisation-2024>

Mura, P. O. & Donath, L. E. (2023). *Digitalisation and Economic Growth in the European Union*. Electronics 12(7). <https://www.mdpi.com/2079-9292/12/7/1718>

Aleca, O. E. & Mihai, F. (2025). *The Role of Digital Infrastructure and Skills in Enhancing Labor Productivity: Insights from Industry 4.0 in the European Union*. Systems 13(2).
<https://www.mdpi.com/2079-8954/13/2/113>

Galik, A. (2021). *The Impact of The Internet on Economic Growth In UE15 (2006-2020) – A Panel Data Analysis*. Economic Policy. <https://ibimapublishing.com/p-articles/38ECO/2021/38133921/>

Nowakowski, W. (2024). *Evaluation of ICT Sector Contributions to Economic Growth*. Journal of Eastern Europe Research in Business and Economics 2024.
<https://ibimapublishing.com/articles/JEERBE/2024/973690/>

Hagsten, E. & Sabadash, A. (2014). *The Impact of Highly-skilled ICT Labour on Firm Performance: Empirical Evidence from Six Countries*. Joint Research Centre.
<https://hdl.handle.net/10419/202193>

Mićić, L. (2017). *Digital transformation and its influence on GDP*. Economics - Innovative and Economics Research Journal 5(1), pp. 1-10. <https://reference-global.com/article/10.1515/eoik-2017-0028>

(2024). *Internet Usage Statistics – Forbes Home*. Forbes. <https://www.forbes.com/home-improvement/internet/internet-statistics/>

Doroiman, M. M. & Sîrghi, N. (2025). *The Impact of Digitalization on EU Competitiveness and Economic Growth: An Econometric Perspective*. The Journal of Economic Behavior. <https://reference-global.com/article/10.2478/tjeb-2025-0006>

Appendix

GDP (\$)	Column1	Internet Users %	Column2	Labor force with advanced education %	Column3	Medium and high-tech exports %	Column4	Avg. of ICT goods & service exports	Column5
Mean	4.3408E+11	Mean	63.32249462	Mean	78.79687578	Mean	47.86984195	Mean	7.762385327
Standard Error	24949923016	Standard Error	0.878503141	Standard Error	0.155373793	Standard Error	0.522715949	Standard Error	0.191759769
Median	1.61185E+11	Median	70.1248	Median	79.405	Median	48.8902092	Median	6.072453866
Mode	#N/A	Mode	57.87157143	Mode	83.311625	Mode	22.34922338	Mode	2.78890704
Standard Deviation	7.58822E+11	Standard Deviation	26.71862995	Standard Deviation	4.725509428	Standard Deviation	15.89778494	Standard Deviation	5.832145691
Sample Variance	5.75811E+23	Sample Variance	713.8851862	Sample Variance	22.33043935	Sample Variance	252.7395661	Sample Variance	34.01392337
Kurtosis	9.010172031	Kurtosis	-0.61559893	Kurtosis	0.53538373	Kurtosis	-0.277016097	Kurtosis	5.802243368
Skewness	2.930452941	Skewness	-0.676403605	Skewness	-0.040789561	Skewness	-0.365753642	Skewness	2.108207405
Range	4.65895E+12	Range	99.716003	Range	32.741	Range	77.53790442	Range	38.95857467
Minimum	984293043.8	Minimum	0.114097	Minimum	62.126	Minimum	3.556592714	Minimum	0.764168329
Maximum	4.65993E+12	Maximum	99.8301	Maximum	94.867	Maximum	81.09449714	Maximum	39.722743
Sum	4.01524E+14	Sum	58573.30753	Sum	72887.1101	Sum	44279.6038	Sum	7180.206428
Count	925	Count	925	Count	925	Count	925	Count	925

Figure 1.1 - Descriptive Statistics

Time	Country Name	Country Code	ICT goods exports %	ICT service exports %	Internet Users %	GDP (\$)	Labor force with advanced education %	Labor force participation rate %	Medium and high-tech exports %	Digital Group	Avg. of ICT goods & service exports	Log_GDP
2000	Albania	ALB	0.39	4.550402787	0.114097	3584570165	75.15442105	66.624	6.559917058	Low	2.470201393	21.99990441
2000	Austria	AUT	6.16	9.462205914	33.7301	1.96182E+11	77.893	71.154	58.89333904	High	7.811102957	26.0023066
2000	Belarus	BLR	1.38	5.234567901	1.8604	12736780455	83.311625	70.583	45.16456985	Low	3.307283951	23.26775974
2000	Belgium	BEL	6.13	9.833528682	29.4317	2.36792E+11	81.567	65.225	53.73163872	High	7.981764341	26.1904499
2000	Bosnia and Herzegovina	BIH	0.2035	7.778730766	1.08296	5567772769	78.73957895	58.692	31.50926249	Low	3.991115383	22.44026095
2000	Bulgaria	BGR	0.79	2.314808999	5.37092	13245990274	74.641	61.466	26.80932125	Low	1.5524045	23.30696072
2000	Croatia	HRV	1.99	3.068575917	6.64488	22134411297	73.38569565	63.379	42.27813314	Low	2.529287958	23.82099931
2000	Czechia	CZE	4.59	2.021276975	9.78053	62175642238	74.191	71.522	57.49591141	High	3.305638488	24.85322916
2000	Denmark	DNK	7.21	5.713313973	39.1724	1.64044E+11	84.005	79.799	52.68571206	High	6.461656986	25.82339941
2000	Estonia	EST	25.26	2.360946528	28.577	5686579748	82.042	70.353	47.36496445	High	13.81047326	22.46137481
2000	Finland	FIN	23.55	4.87445872	37.2485	1.25968E+11	81.956	74.582	55.28319689	High	14.21222936	25.55929232
2000	France	FRA	10.81	7.166122242	14.3079	1.36096E+12	77.709	68.472	66.88979359	High	8.988061121	27.93921077
2000	Germany	DEU	8.4	6.211465813	30.2163	1.96698E+12	76.435	71.438	72.97261619	High	7.305732907	28.30752084
2000	Greece	GRC	4.25	1.753885633	9.13884	1.2576E+11	81.624	63.182	27.92383583	Low	3.001942817	25.55764249
2000	Hungary	HUN	25.74	2.549350957	6.99968	47275954429	76.634	62.206	73.33582314	High	14.14467548	24.57926764
2000	Iceland	ISL	0.11	6.507967334	44.4705	9025660362	94.867	88.549	25.89155318	High	3.308983667	22.92333751
2000	Ireland	IRL	36.5	42.94548599	17.8505	1.00208E+11	86.279	73.834	58.02576806	High	39.722743	25.3050998
2000	Italy	ITA	4.49	5.824788942	23.1109	1.14966E+12	79.652	59.895	52.87932896	Low	5.157394471	27.77048855
2000	Latvia	LVA	0.87	11.09490773	6.31906	7761252607	77.691	67.152	15.47920439	High	5.982453866	22.77240958
2000	Lithuania	LTU	4.77	3.638373229	6.42707	11550695727	81.4	71.111	32.13351445	Low	4.204186615	23.17001151
2000	Luxembourg	LUX	11.23	4.213025631	22.8873	21230182989	76.706	64.266	42.53627914	High	7.721512815	23.77868973
2000	Malta	MLT	63.64	2.491406788	13.1137	4036809865	85.714	58.181	81.09449714	Low	33.06570339	22.11872058
2000	Moldova	MDA	0.7	8.681759099	1.28285	1288429392	76.254	65.711	10.68556936	Low	4.690879549	20.97668979
2000	Montenegro	MNE	0.592666667	4.985147412	57.87157143	984293043.8	79.68530769	56.574	22.34922338	Low	2.78890704	20.70743422
2000	Netherlands	NLD	17.88	5.230582951	43.9844	4.17649E+11	80.171	80.554	60.85302722	High	11.55529148	26.75790788
2000	North Macedonia	MKD	0.29	9.031479398	2.48557	3772859034	80.23178947	59.788	16.05040503	Low	4.660739699	22.05109892
2000	Norway	NOR	1.84	4.844134453	52	1.71457E+11	86.432	79.334	47.93770411	High	3.342067226	25.86759952
2000	Poland	POL	4.1	2.842825479	7.28543	1.72954E+11	79.571	65.166	47.79289148	Low	3.47141274	25.87628877
2000	Portugal	PRT	6.12	2.419055133	16.4305	1.18605E+11	87.747	71.031	43.22690944	Low	4.269527567	25.49906611
2000	Romania	ROU	4.99	10.72644722	3.61372	37253739511	78.908	56.801	26.3819753	Low	7.85822361	24.34101817
2000	Serbia	SRB	1.521333333	16.12982427	57.15133333	7326373882	69.78594444	64.939	28.33792874	Low	8.825578804	22.71474653
2000	Slovenia	SVN	1.94	4.049118721	15.1103	20159190702	77.967	67.508	54.29439668	High	2.994559361	23.72692614

Figure 1.2 - Data Cleaning & Log Calculation

t-Test: Two-Sample Assuming Unequal Variances

	Variable 1	Variable 2
Mean	6.22E+11	2.35909E+11
Variance	8.37E+23	2.24772E+23
Observations	475	450
Hypothesized Mean Difference	0	
df	720	
t Stat	8.115052	
P(T<=t) one-tail	1.05E-15	
t Critical one-tail	1.646973	
P(T<=t) two-tail	2.1E-15	
t Critical two-tail	1.963264	

Figure 2.1 - t-Test Results

SUMMARY OUTPUT

Regression Statistics								
Multiple R	0.669647169							
R Square	0.448427331							
Adjusted R Square	0.446029189							
Standard Error	1.340735654							
Observations	925							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	4	1344.508278	336.1270695	186.9894792	2.8432E-117			
Residual	920	1653.766325	1.797572093					
Total	924	2998.274603						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	25.74524441	0.74634822	34.49494984	5.3178E-168	24.28050179	27.20998703	24.28050179	27.20998703
Internet Users %	0.012585229	0.001803909	6.976644088	5.7764E-12	0.009044975	0.016125482	0.009044975	0.016125482
Labor force with advanced education %	-0.051076124	0.009519456	-5.365445505	1.02223E-07	-0.069758493	-0.032393754	-0.069758493	-0.032393754
Avg. of ICT goods & service exports	-0.057162174	0.00821193	-6.960869774	6.42541E-12	-0.073278463	-0.041045885	-0.073278463	-0.041045885
Medium and high-tech exports %	0.071333526	0.003186924	22.38319102	6.21656E-89	0.065079042	0.07758801	0.065079042	0.07758801

Figure 3.1 - Regression Summary Output

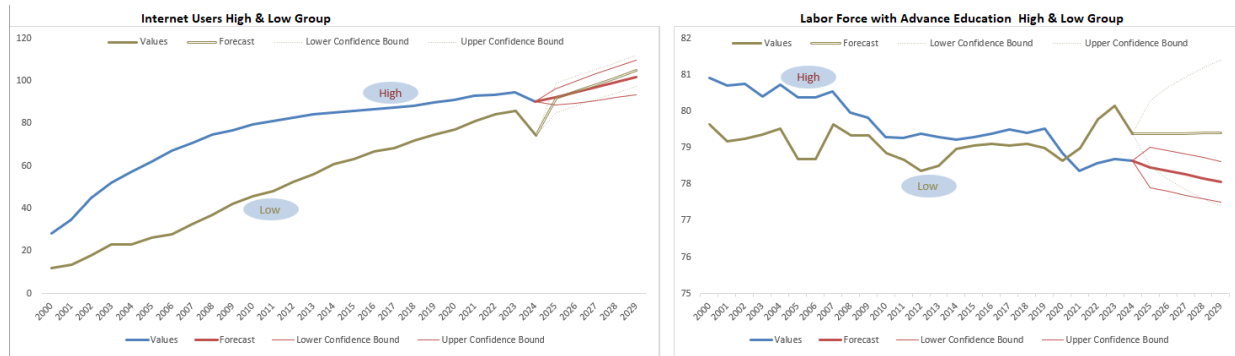


Figure 3.2.1 - Forecast Graphs

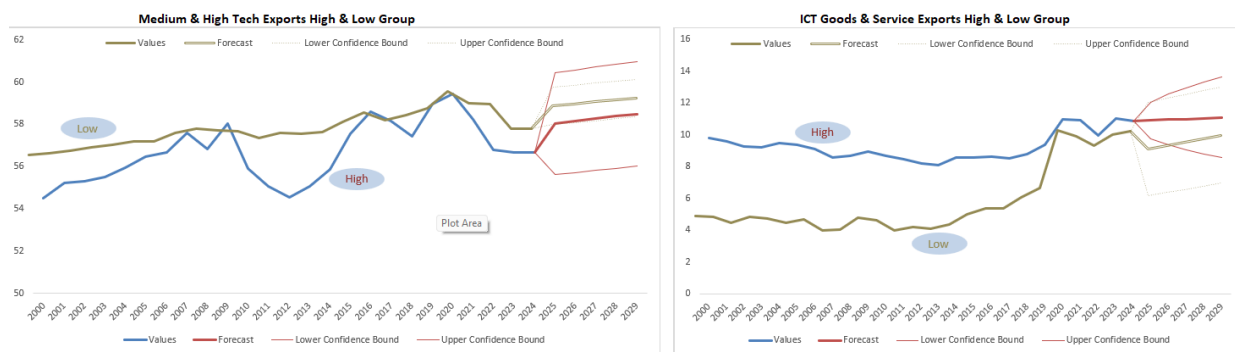


Figure 3.2.2 - Forecast Graphs

Scenario Summary				
Current Values:		Status Quo	Digital Investment Strategy +5%	Aggressive Growth +10%
Changing Cells:				
\$C\$3	63.32	63.32	66.49	69.65
\$C\$4	78.80	78.80	79.78	80.77
\$C\$5	7.76	7.76	8.15	8.54
\$C\$6	47.87	47.87	50.26	52.66
Result Cells:				
\$B\$8	25.49	25.49	25.63	25.76
\$B\$9	117363680477.52	117363680477.52	134729437997.75	154683296793.24

Notes: Current Values column represents values of changing cells at time Scenario Summary Report was created. Changing cells for each scenario are highlighted in gray.

Figure 4.1 - Scenario Summary Report

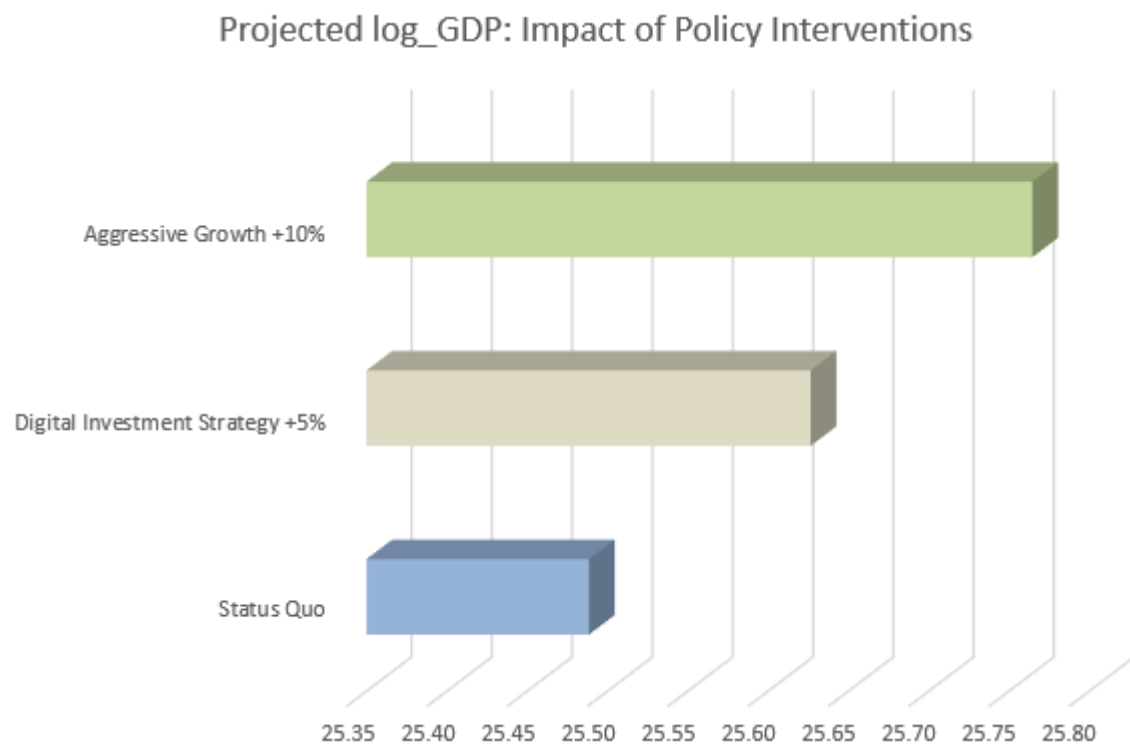


Figure 4.2 - Visual representation of Scenario Impacts

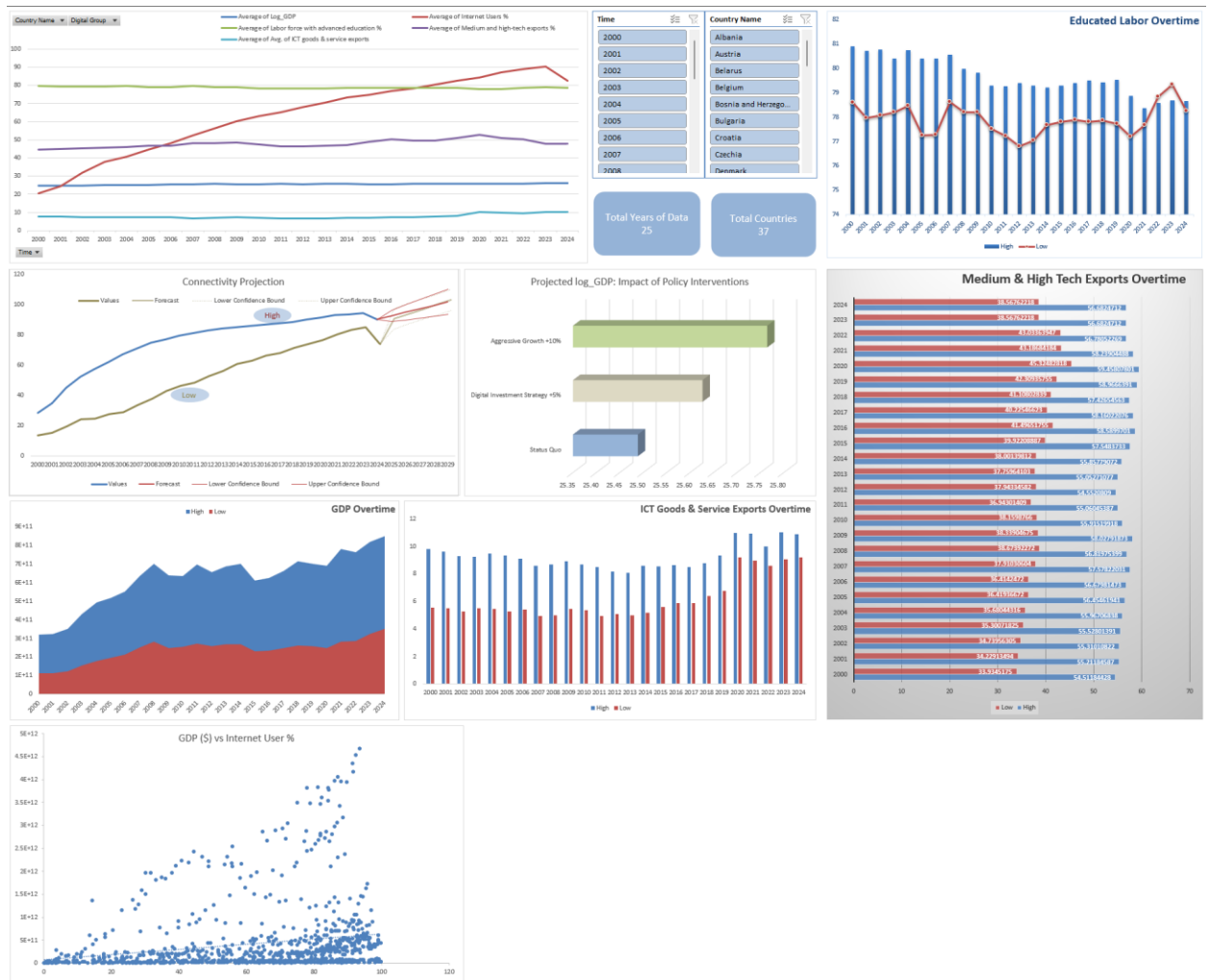


Figure 5.1 - Full Dashboard

	A	B	C	D	E	F	G	H	I	J	K	L
1	Time	Country Name	Country Code	ICT goods exports (% of total goods exports)	ICT service exports (% of service exports, BoP)	Individuals using the Internet (% of population)	GDP (current US\$)	Labor force with advanced education (% of total working-age population with advanced education)	Labor force participation rate, total (% of total population ages 15-64) (modeled ILO estimate)	Medium and high-tech exports (% manufactured exports)		
2	2000	Albania	ALB	0.39	4.550402787	0.114097	3584570165 ..			66.624	6.559917058	
3	2000	Andorra	AND	8.07 ..		10.5388	1432606189 ..		..			
4	2000	Austria	AUT	6.16 ..		33.7301	1.96182E+11	77.893	71.154	58.89333904		
5	2000	Belarus	BLR	1.38	5.234567901	1.8604	12736780455 ..			70.583	45.16456985	
6	2000	Belgium	BEL	6.13 ..		29.4317	2.36792E+11	81.567	65.225	53.73163872		
7	2000	Bosnia and Herzegovina	BIH	..	7.778730766	1.08296	5567772769 ..			58.692	31.50926249	
8	2000	Bulgaria	BGR	0.79	2.314808999	5.37092	13245990274	74.641	61.466	26.80932125		
9	2000	Croatia	HRV	1.99	3.068575917	6.64488	22134411297 ..			63.379	42.27813314	
10	2000	Czechia	CZE	4.59	2.021276975	9.78053	62175642238	74.191	71.522	57.49591141		
11	2000	Denmark	DNK	7.21 ..		39.1724	1.64044E+11	84.005	79.799	52.68571206		
12	2000	Estonia	EST	25.26	2.360946528	28.577	5686579748	82.042	70.353	47.36496445		
13	2000	Finland	FIN	23.55	4.87445872	37.2485	1.25968E+11	81.956	74.582	55.28319689		
14	2000	France	FRA	10.81	7.166122242	14.3079	1.36096E+12	77.709	68.472	66.88979359		
15	2000	Germany	DEU	8.4	6.211465813	30.2163	1.96698E+12	76.435	71.438	72.97261619		
16	2000	Greece	GRC	4.25	1.753885633	9.13884	1.2576E+11	81.624	63.182	27.92383583		
17	2000	Hungary	HUN	25.74	2.549350957	6.99968	47275954429	76.634	62.206	73.33582314		
18	2000	Iceland	ISL	0.11 ..		44.4705	9025660362	94.867	88.549	25.89155318		
19	2000	Ireland	IRL	36.5 ..		17.8505	1.00208E+11	86.279	73.834	58.02576806		
20	2000	Italy	ITA	4.49	5.824788942	23.1109	1.14966E+12	79.652	59.895	52.87932896		
21	2000	Kosovo	XKX	..	..	..	..	83.05 ..	..	..		
22	2000	Latvia	LVA	0.87 ..		6.31906	7761252607	77.691	67.152	15.47920439		
23	2000	Liechtenstein	LIE	..		36.5152	2483889858 ..	..	..	..		
24	2000	Lithuania	LTU	4.77	3.638373229	6.42707	11550695727	81.4	71.111	32.13351445		
25	2000	Luxembourg	LUX	11.23	4.213025631	22.8873	21230182989	76.706	64.266	42.53627914		
26	2000	Malta	MLT	63.64	2.491406788	13.1137	4036809865	85.714	58.181	81.09449714		
27	2000	Moldova	MDA	0.7	8.681759099	1.28285	1288429392	76.254	65.711	10.68556936		
28	2000	Montenegro	MNE	..	..	..	984293043.8 ..		56.574	22.34922338		
29	2000	Monaco	MCO	..	..	42.1849	2654462665 ..	..	..	..		
30	2000	Netherlands	NLD	17.88	5.230582951	43.9844	4.17649E+11	80.171	80.554	60.85302722		
31	2000	North Macedonia	MKD	0.29	9.031479398	2.48557	3772859034 ..		59.788	16.05040503		
32	2000	Norway	NOR	1.84	4.844134453	52	1.71457E+11	86.432	79.334	47.93770411		

< > Original Data Series - Metadata Task1-D Means Task2-D pivotTables Descriptive stats t-Test Dashboard Forecast Task3-D Regression Output Task4 Scenario Summary +

Snap of Excel Workbook

Formulas used

For Mean imputation country-wise

=IF(F2="",XLOOKUP(B2,Means!\$A\$2:\$A\$38,Means!\$C\$2:\$C\$38),F2)

For Digital Group

=XLOOKUP(B3,Means!\$A\$2:\$A\$38,Means!\$J\$2:\$J\$38)